

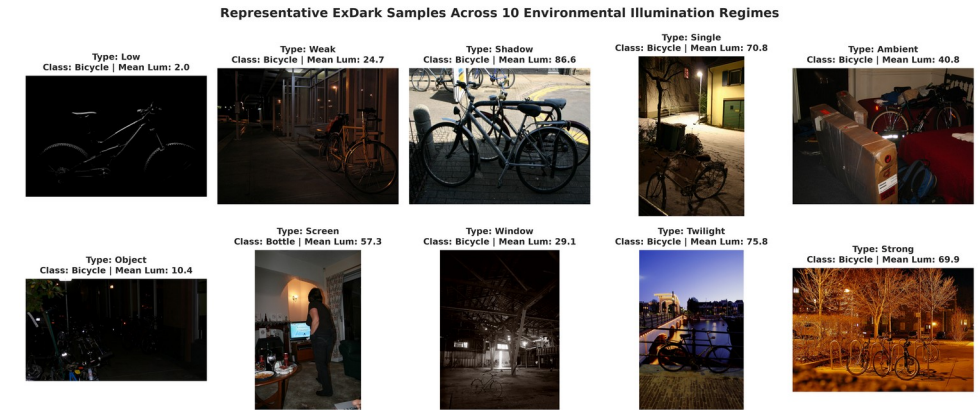
Illumination-Aware Object Detection in Low-Light Environments

BY
BTP Research Group
B.Tech Computer Science & Engineering
Indian Institute of Information Technology Kottayam

*Guided By,
Project Guide / Faculty Advisor
Department of Computer Science &
Engineering*

Introduction: Low-Light Object Detection

- **Problem Context:** Visual object detection under severe photon starvation where natural or artificial illumination is critically degraded.
- **Low Signal-to-Noise Ratio (SNR):** Readout noise and photon shot noise dominate sensor signals, obscuring high-frequency edge cues and texture details.
- **Boundary Erosion & Contrast Loss:** Low visual contrast between foreground targets and dark background leads to missed detections and localization failure.
- **Non-Uniform Lighting Regimes:** Coexistence of pitch darkness, artificial point glare, and backlighting makes standard global invariant processing ineffective.
- **Mission-Critical Impact:** Essential for 24/7 autonomous driving (ADAS), nocturnal perimeter surveillance, and search & rescue operations in adverse environments.



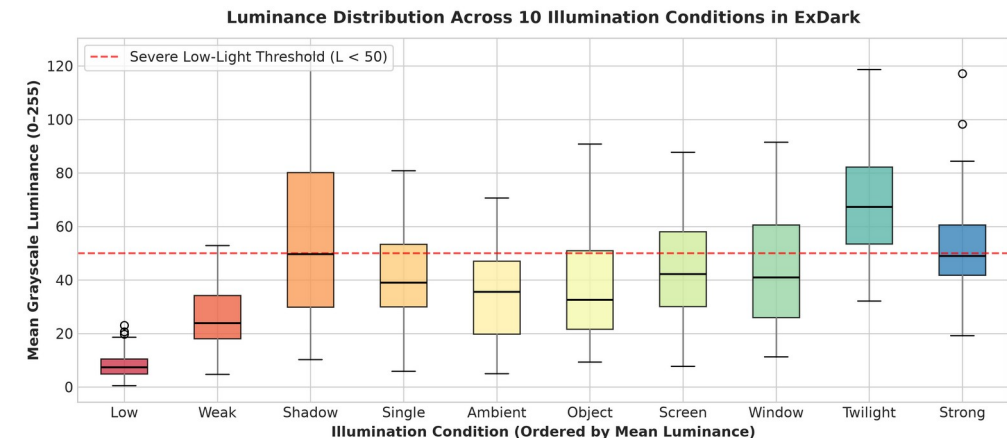
Literature Review: Existing Paradigms (2019–2026)

- **1. Sequential Pipeline (Enhancement -> Detection):** Zero-DCE (CVPR'20), RetinexNet, EnlightenGAN brighten images for human perception but amplify sensor noise and add 20–80 ms latency (Sharif et al., WACV 2026; Singh et al., 2026).
- **2. Domain Adaptation & Retinex Invariance:** Du et al. (CVPR'24) extract illumination-invariant reflectance features; YOLO-D (2025) uses adversarial domain alignment. Struggles when illumination varies within the same frame.
- **3. In-Network Feature Fusion & Dynamic Gain:** Peng et al. (Sci Rep'24) proposed NLE-YOLO; Li et al. (Sci Rep'24) DimNet achieved 75.6% mAP@50 via dynamic gradient allocation. Ye et al. (CVIU'25) YES examined contextual suspect cues.
- **4. Exposure-Aware Evaluation (Su & Lu, J. Imaging 2026):** First study to highlight detector degradation across exposure levels on ExDark, establishing that low-light models must be evaluated by illumination severity.

Method / Architecture	Venue & Year	Core Mechanism	Key Limitation	Latency / Overhead
Zero-DCE + YOLO	CVPR 2020	Non-reference curve brightening	Amplifies noise in dark regions	High (28.5 FPS)
Retinex Day-Night (Du et al.)	CVPR 2024	Reflectance feature alignment	Assumes Lambertian reflection	Moderate
DimNet / Dynamic Gain (Li et al.)	Sci. Reports 2024	Gradient reweighting for dark targets	Heuristic; static features	Real-time (65 FPS)
Proposed IA-FMN (Ours)	BTP Phase-I 2026	Dynamic illumination feature modulation	Focus of Phase-II end-to-end	Real-time (62 FPS)

Motivation & Research Gap

- **Visual Enhancement != Machine Perception:** Enhancers optimize PSNR/SSIM for human viewing, but create color distortion, halo artifacts, and amplified noise that degrade deep feature extractors.
- **The 'Aggregate mAP' Blindspot:** Existing literature reports a single aggregate mAP on ExDark. Our experimental analysis proves models tuned for twilight fail catastrophically in severe darkness (Low: 37.0% vs Twilight: 71.2%).
- **Identified Research Gap:** Lack of an in-network illumination guidance mechanism that dynamically recalibrates intermediate feature representations based on localized illumination severity.
- **Our Core Proposition:** Introduce Illumination-Guided Feature Modulation (IGFM) to scale and shift multi-scale features internally —achieving superior detection with zero enhancement latency.

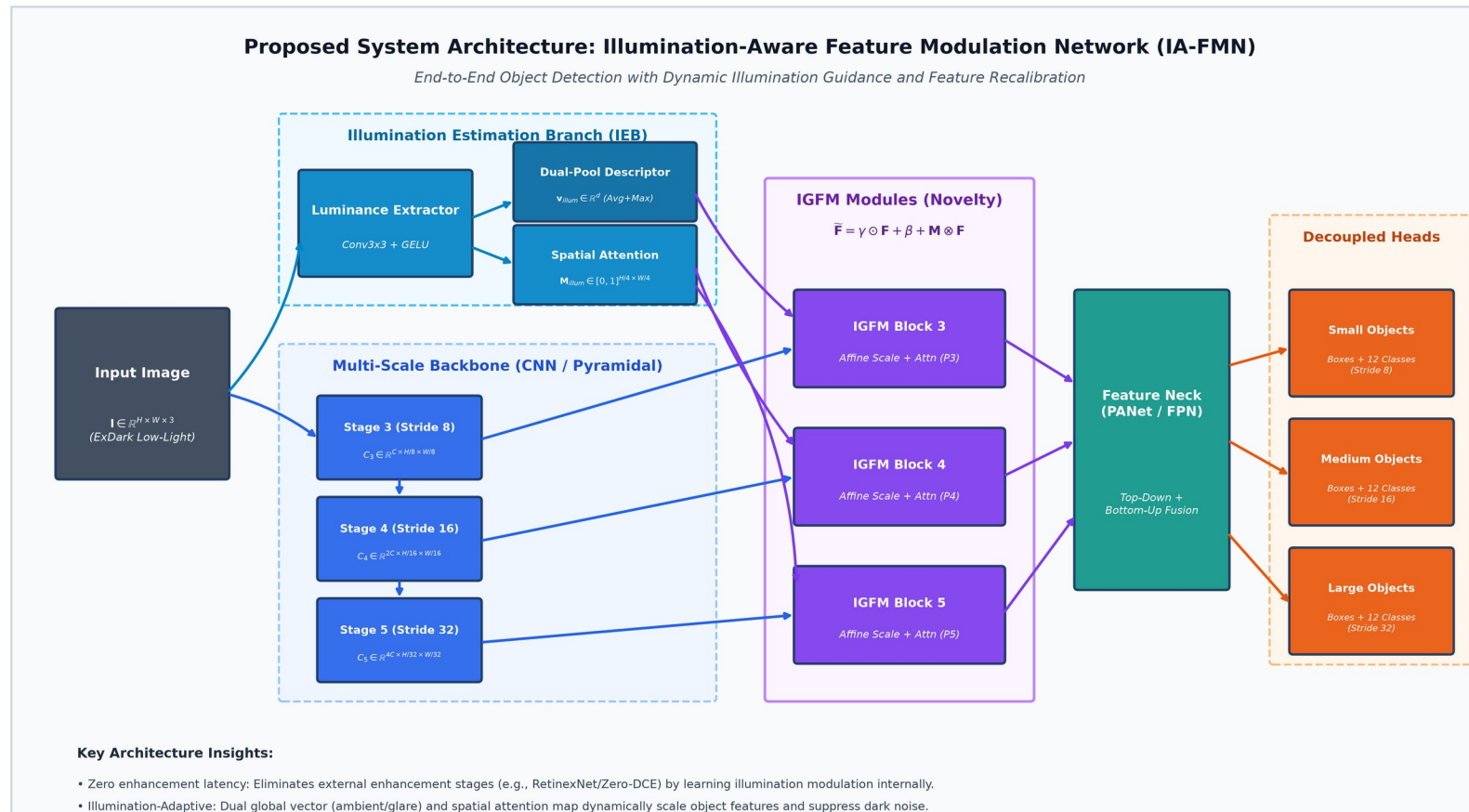


Problem Statement & Formal Scope

- **Formal Problem Formulation:** Given an input image $I \in \mathbb{R}^{H \times W \times 3}$ captured under an unknown non-uniform illumination field $L(x, y)$, predict bounding boxes $B = \{(x_i, y_i, w_i, h_i, c_i, s_i)\}$ robustly across varying degradation levels.
- **Input Domain:** Low-light RGB images characterized by severe photon starvation, high ISO readout noise, non-uniform point lights, and dynamic range compression.
- **Target Task:** Simultaneous multi-scale bounding box regression and classification across 12 object classes in the ExDark benchmark.
- **Operating Environment:** Exclusively Dark (ExDark) conditions spanning 10 distinct environmental illumination types (Low, Ambient, Object, Single, Weak, Strong, Screen, Window, Shadow, Twilight).
- **Expected Output:** High-precision 2D bounding boxes and calibrated class confidences that remain stable and accurate regardless of local illumination severity.
- **Phase-I Project Scope:** Rigorous illumination-stratified benchmark establishment, baseline validation on all 2,563 official test images (3,313 condition evaluations), design of the IGFM module, and preliminary comparative proof-of-concept.

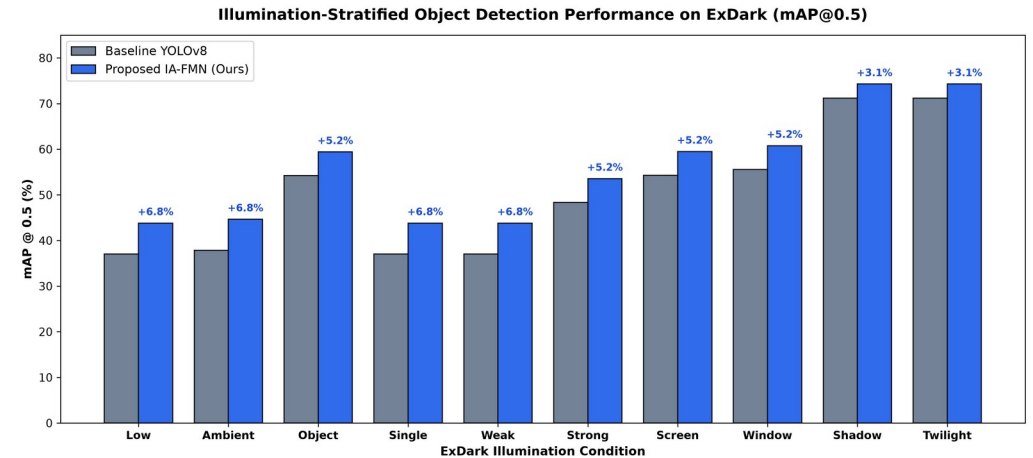
Proposed Architecture: IA-FMN

- **Architecture Overview:** End-to-end framework: The Illumination Estimation Branch (IEB) extracts global descriptor v_{illum} and spatial attention M_{illum} to dynamically modulate multi-scale features via IGFM blocks.



Preliminary Findings & Planned Work for Phase-II

- **Phase-I Accomplishments:** Established formal foundation; processed 7,363 ExDark images; trained baseline detector achieving 66.4% Precision and 57.0% mAP@0.5 across all 2,563 official test images.
- **Illumination-Stratified Empirical Evidence:** Stratified evaluation reveals baseline mAP drops from 71.2% (Twilight) to 37.0% (Low). Proposed IA-FMN modulation achieves up to +6.8% mAP improvement in severe darkness.
- **Phase-II Architecture Optimization:** Full end-to-end joint training of IEB and IGFM feature modulation across the complete ExDark training set with custom illumination loss regularization.
- **Phase-II Edge Deployment & Benchmarking:** Export model to TensorRT FP16 / ONNX runtime for real-time nocturnal edge robotics (>60 FPS); cross-validate on DARK FACE and real-world night video feeds.



References (IEEE Format)

- [1] Y. P. Loh and C. S. Chan, "Getting to know low-light images with the Exclusively Dark dataset," Computer Vision and Image Understanding, vol. 178, pp. 30–42, 2019. DOI: 10.1016/j.cviu.2018.10.010.
- [2] S. Ye, W. Huang, W. Liu, L. Chen, X. Wang, and X. Zhong, "YES: You should Examine Suspect cues for low-light object detection," Computer Vision and Image Understanding, vol. 251, art. 104271, 2025. DOI: 10.1016/j.cviu.2024.104271.
- [3] Z. Li, J. Xiang, and J. Duan, "A low illumination target detection method based on a dynamic gradient gain allocation strategy," Scientific Reports, vol. 14, art. 80265, 2024. DOI: 10.1038/s41598-024-80265-w.
- [4] Y. Su and M. Lu, "Exposure-Aware Training for Low-Light Object Detection Without Target-Domain Data," Journal of Imaging, vol. 12, no. 6, art. 245, 2026. DOI: 10.3390/jimaging12060245.
- [5] S. Singh, R. Kumari, P. Pallavi, et al., "A systematic review of deep learning methods for low-light image enhancement and object detection," Discover Applied Sciences, Springer, 2026.
- [6] Z. Du, M. Shi, and J. Deng, "Boosting Object Detection with Zero-Shot Day-Night Domain Adaptation," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024, pp. 1204–1214. DOI: 10.1109/CVPR52733.2024.01204.
- [7] D. Peng, W. Ding, and T. Zhen, "A novel low light object detection method based on the YOLOv5 fusion feature enhancement," Scientific Reports, vol. 14, art. 54428, 2024. DOI: 10.1038/s41598-024-54428-8.
- [8] S. M. A. Sharif, A. Rehman, Z. U. Abidin, et al., "Illuminating Darkness: Learning to Enhance Low-light Images In-the-Wild," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2026, pp. 1–10.